Reward Support Across Episode Positions: Exact Counts and Their Effect on GAE Critic Targets

Technical analysis for the tokamak control experiments

October 26, 2026

Abstract

A fixed-length rollout generally ends at a different position in each episode. Consequently, advantages associated with different episode positions use different numbers of observed rewards. We derive the exact cumulative support count C_t, its periodic structure, and its relation to critic-target error. For the example $M = 500$, $N = 256$, and $m = 64$, the first position uses 130 rewards per target on average, whereas the final position uses one. Under a controlled endpoint-error perturbation, their average sensitivity coefficients are approximately 0.0651 and 0.995, respectively. All results assume **correct bootstrapping at artificial episode time limits**. The analysis separates reward support, sample count, target bias, and state coverage so that the effect can be quantified without treating these different quantities as interchangeable.

1 Sampling convention and the quantities being counted

The sampling convention is the one illustrated by Figure 14.1 of the accompanying *Tokamak Plasma Simulator Guide*: rollout windows continue across resets, while GAE recursion stops at both rollout boundaries and episode boundaries. The derivation here is self-contained and does not require that figure.

Consider one environment. It produces m episodes of exactly M action steps each, with no early failures. Rollouts contain N transitions. The first rollout starts at the first transition of the first episode. Episode j is indexed by $j = 1, \ldots, m$, and position within an episode by $t = 1, \ldots, M$. At position t, the environment starts in $s_{j,t}$, takes $a_{j,t}$, receives $r_{j,t}$, and reaches $s_{j,t+1}$.

An artificial time limit resets the environment after transition M but does not make the underlying control task terminal. Its final state is used for bootstrap; rewards from the reset episode are never appended to the old episode's GAE. A final partial rollout is assumed to be processed, or sampling is continued until the containing rollout is complete. Discarding an incomplete final rollout would require excluding the discarded samples from the counts below.

Table 1: Definitions. Counts include the current transition's reward.

Symbol	Meaning
m	Number of complete episodes from one environment.
M	Actual episode length in transitions, not merely a configured maximum.
N	Rollout length per environment.
$K_{j,t}$	Number of rewards available to the GAE fragment starting at (j, t).
C_t	Sum of those support lengths over all m episodes.
$\bar{K}_t = C_t/m$	Average number of available rewards per target at position t.
γ, λ	Discount and GAE trace parameters; $q = \gamma\lambda$.

There are m training samples at every position t, regardless of C_t. C_t counts reward occurrences supporting these targets, not independent samples and not additional simulator calls. Counts for different positions can reuse the same rewards. A bootstrap value is a prediction and is not counted as an observed reward. For $0 < \gamma\lambda < 1$, every available reward has a nonzero coefficient; when $\lambda = 0$, only the current reward contributes despite the larger available support.

2 Exact reward-support count

2.1 One sample

The zero-based global transition index of (j, t) is

$$g_{j,t} = (j-1)M + t - 1. \tag{2.1}$$

The number of transitions remaining in its rollout, including the current one, is

$$B_{j,t} = N - (g_{j,t} \bmod N). \tag{2.2}$$

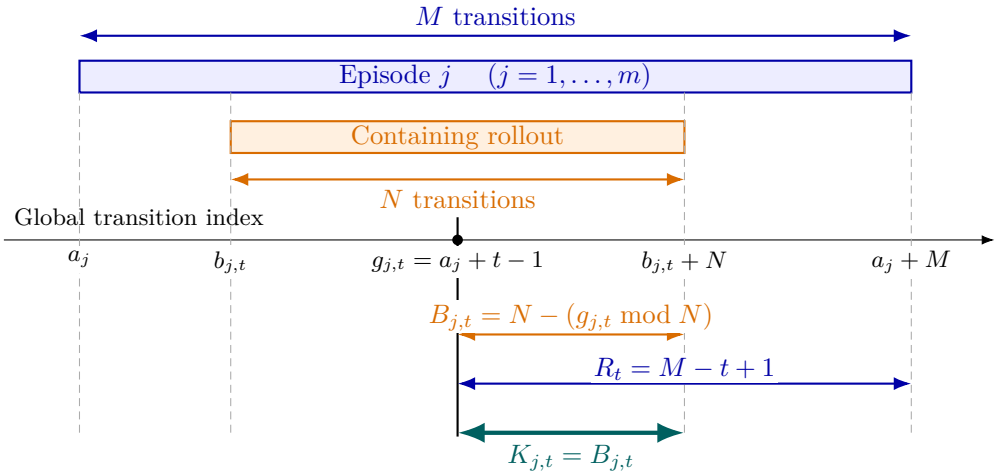
The number remaining in its episode is $R_t = M - t + 1$. Therefore,

$$\boxed{K_{j,t} = \min\left\{M - t + 1,\ N - \left(((j-1)M + t - 1) \bmod N\right)\right\}.} \tag{2.3}$$

The actual rewards used are $r_{j,t}, \ldots, r_{j,t+K_{j,t}-1}$. Both boundaries matter: continuing to fill a rollout after reset does not extend the preceding episode's reward support.

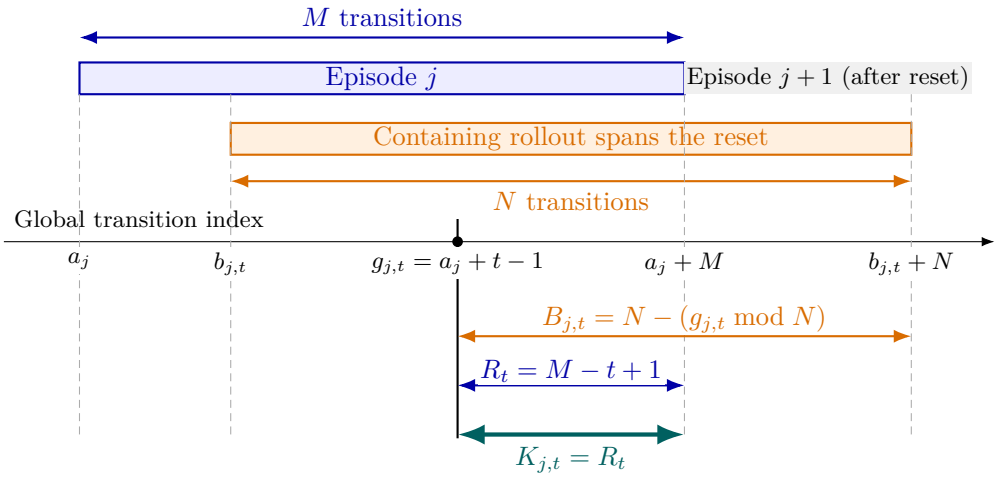
Figure 1 makes the endpoints of these counts explicit. Let $a_j = (j-1)M$ be the episode-start index and $b_{j,t} = N\lfloor g_{j,t}/N \rfloor$ the containing rollout's start index. The episode occupies $[a_j, a_j + M)$ and the rollout occupies $[b_{j,t}, b_{j,t} + N)$. Their common right endpoint for this target is $h_{j,t} = \min(a_j + M, b_{j,t} + N)$, so $K_{j,t} = h_{j,t} - g_{j,t}$. These are half-open intervals of *transition indices*: the endpoint is where a bootstrap state is available, not an extra observed reward.

(a) Rollout ends first: $B_{j,t} < R_t$



Bootstrap at $s_{j,t+K_{j,t}}$; the next rollout can continue episode j.

(b) Episode ends first: $R_t < B_{j,t}$



Bootstrap at $s_{j,M+1}$, before reset; episode $j+1$ rewards are excluded.

$$K_{j,t} = \min\{B_{j,t}, R_t\} \quad \text{observed rewards: } r_{j,t}, \ldots, r_{j,t+K_{j,t}-1}$$

$$C_t(m; M, N) = \sum_{j=1}^{m} K_{j,t} \qquad \bar{K}_t = C_t/m$$

Figure 1: Reading the reward-support intervals. Blue denotes episode spans, orange denotes rollout spans, and teal denotes the usable reward fragment. Every double-headed arrow measures a number of transitions between its two endpoints, including the current transition and excluding the right endpoint. The global index $g_{j,t}$ counts transitions from the start of sampling; t is the one-based position within episode j. Panels are schematic and independently scaled, not numerical examples with a shared scale. If both boundaries coincide, $K_{j,t} = B_{j,t} = R_t$ and the episode-boundary bootstrap rule applies. The bootstrap value is not counted in $K_{j,t}$ or C_t.

2.2 All episodes at the same position

Summing Equation (2.3) gives the desired exact count:

$$\boxed{C_t(m; M, N) = \sum_{j=1}^{m} \min\left\{M - t + 1,\ N - \big(((j-1)M + t - 1) \bmod N\big)\right\}.} \tag{2.4}$$

Consequently,

$$m \le C_t \le m \min(N, M - t + 1), \qquad C_M = m. \tag{2.5}$$

The final position always has one observed reward per target. More episodes increase the number of such targets but do not extend any individual target's observed future beyond the episode boundary.

3 Periodicity and a closed form

Let

$$d = \gcd(M, N), \qquad L = \frac{N}{d}. \tag{3.1}$$

Episode starts advance the rollout phase by $M \bmod N$. After L episodes, LM is divisible by N, so $K_{j+L,t} = K_{j,t}$. Write $m = uL + b$, with $u = \lfloor m/L \rfloor$ and $0 \le b < L$. Then

$$C_t = uS_t + \sum_{j=1}^{b} K_{j,t}, \qquad S_t = \sum_{j=1}^{L} K_{j,t}. \tag{3.2}$$

For a fixed t, define

$$a_t = (t-1) \bmod d, \qquad c_t = d - a_t. \tag{3.3}$$

Over one period, the rollout residual lengths visit exactly the set

$$\{c_t, c_t + d, \ldots, c_t + (L-1)d\}. \tag{3.4}$$

Their chronological order depends on M, but the full-period sum does not. Let h_t count the members no larger than R_t:

$$h_t = \min\left\{L, \max\left(0, \left\lfloor \frac{R_t - c_t}{d} \right\rfloor + 1\right)\right\}. \tag{3.5}$$

Summing the arithmetic progression below R_t and clipping the remaining terms to R_t gives

$$\boxed{S_t = h_t c_t + \frac{d\, h_t(h_t - 1)}{2} + (L - h_t)R_t.} \tag{3.6}$$

Equations (3.2)–(3.6) give a closed form for complete periods plus an exact remainder for any finite m.

3.1 Two useful special cases

If N divides M, episode starts always have the same rollout phase:

$$C_t = m \min\{M - t + 1,\ N - ((t-1) \bmod N)\}. \tag{3.7}$$

Repeating episodes does not average over different phases in this case.

If $\gcd(M, N) = 1$ and m is a multiple of N, every residual length from 1 to N occurs equally often. With $k_t = \min(M - t + 1, N)$,

$$\boxed{\bar{K}_t = k_t - \frac{k_t(k_t - 1)}{2N}.} \tag{3.8}$$

Far enough from the episode boundary, $\bar{K}_t = (N+1)/2$, rather than N. For $d > 1$, residual lengths occupy a residue class; this can produce small position-dependent oscillations even far from the episode endpoint.

4 Numerical example: 500-step episodes and 256-step rollouts

Take $M = 500$, $N = 256$, and $m = 64$. Here $d = 4$ and $L = 64$, so the episodes cover one full phase period: $64 \times 500 = 125 \times 256 = 32000$ transitions.

Table 2: Exact reward support over a complete phase period.

t	$K_{1,t}$	$K_{2,t}$	C_t	$\bar{K}_t$
1	256	12	8320	130.0000
100	157	169	8128	127.0000
256	1	13	8116	126.8125
257	244	12	8296	129.6250
400	101	101	5164	80.6875
450	51	51	2952	46.1250
499	2	2	128	2.0000
500	1	1	64	1.0000

The first rollout contains episode 1, steps 1–256. The next contains episode 1, steps 257–500, followed by episode 2, steps 1–12. Thus the first position of episode 2 has only 12 rewards available, while the first position of episode 1 has 256. Phase averaging redistributes rollout-edge effects, but it cannot remove the episode-end constraint: the final position always has support one.

5 How observed rewards enter a critic target

5.1 A fixed-policy estimation problem

Fix a policy π and an old critic V_{old} while constructing targets. Assume the state is Markov; if the task has a time-varying reference, the state must contain the information needed to predict that reference. The continuing, discounted value is

$$V^\pi(s) = \mathbb{E}_\pi\left[\sum_{\ell=0}^{\infty} \gamma^\ell r_{t+\ell} \mid s_t = s\right], \qquad e(s) = V_{\text{old}}(s) - V^\pi(s). \tag{5.1}$$

All rewards and values are on the same scale. If reward normalization is used, freeze its statistics for the diagnostic experiment.

A squared-error critic regression fits targets Y_t:

$$\mathcal{L}_V(\phi) = \frac{1}{2}\mathbb{E}[(V_\phi(s_t) - Y_t)^2]. \tag{5.2}$$

With unlimited conditional data and unrestricted function fitting, its minimizer is $\mathbb{E}[Y_t \mid s_t = s]$. Thus a biased target mean is directly a biased regression objective. Finite data, shared-network approximation, and incomplete optimization add further error; a coefficient derived below describes target sensitivity, not a guarantee of the eventual network error.

5.2 The simpler K-step target

For the bootstrapped K-step return,

$$Y_t^{(K)} = \sum_{\ell=0}^{K-1} \gamma^\ell r_{t+\ell} + \gamma^K V_{\text{old}}(s_{t+K}), \tag{5.3}$$

Derivation of the target-bias identity. Here $\mathbb{E}[\cdot \mid s]$ abbreviates $\mathbb{E}_\pi[\cdot \mid s_t = s]$: the starting state is fixed, while the current and subsequent actions are sampled from the fixed policy π, and transitions follow the environment dynamics. The old critic is held fixed when taking this expectation. Let K be a fixed positive integer, and assume the discounted returns are integrable; bounded rewards and $0 \le \gamma < 1$ are sufficient.

Step 1: split the actual infinite return at transition K. Define $G_t = \sum_{\ell=0}^{\infty} \gamma^\ell r_{t+\ell}$. Reindexing the tail gives

$$\begin{aligned} G_t &= \sum_{\ell=0}^{K-1} \gamma^\ell r_{t+\ell} + \sum_{\ell=K}^{\infty} \gamma^\ell r_{t+\ell} \\ &= \sum_{\ell=0}^{K-1} \gamma^\ell r_{t+\ell} + \gamma^K \underbrace{\sum_{u=0}^{\infty} \gamma^u r_{t+K+u}}_{G_{t+K}}. \end{aligned}$$

This is an equality for the realized return on a continuing trajectory. The factor γ^K converts a return measured from time $t+K$ into units discounted back to time t.

Step 2: replace the tail by its conditional mean, not by a samplewise equality. Let $\mathcal{H}_{t+K}$ denote the observed trajectory history up to and including s_{t+K}, before choosing a_{t+K}. Under the Markov-state and fixed-policy assumptions,

$$\mathbb{E}_\pi[G_{t+K} \mid \mathcal{H}_{t+K}] = V^\pi(s_{t+K}).$$

Since s_t belongs to this history, the law of iterated expectations gives

$$\begin{aligned} \mathbb{E}_\pi[G_{t+K} \mid s_t = s] &= \mathbb{E}_\pi[\mathbb{E}_\pi[G_{t+K} \mid \mathcal{H}_{t+K}] \mid s_t = s] \\ &= \mathbb{E}_\pi[V^\pi(s_{t+K}) \mid s_t = s]. \end{aligned}$$

Taking conditional expectations in Step 1 therefore yields the K-step Bellman identity:

$$V^\pi(s) = \mathbb{E}_\pi\left[\sum_{\ell=0}^{K-1} \gamma^\ell r_{t+\ell} + \gamma^K V^\pi(s_{t+K}) \,\middle|\, s_t = s \right].$$

The realized tail G_{t+K} need not equal $V^\pi(s_{t+K})$; their conditional means agree. No continuation rollout is required to compute the bootstrap target, even though the true continuation is used to define its correctness.

Step 3: subtract the true-value identity from the mean training target. Both expectations use the same policy, dynamics, starting state, and horizon. By linearity of conditional expectation, the identical reward terms cancel:

$$\begin{aligned} &\mathbb{E}_\pi[Y_t^{(K)} \mid s_t = s] - V^\pi(s) \\ &\quad = \mathbb{E}_\pi\left[\left(\sum_{\ell=0}^{K-1} \gamma^\ell r_{t+\ell} + \gamma^K V_{\text{old}}(s_{t+K}) \right) - \left(\sum_{\ell=0}^{K-1} \gamma^\ell r_{t+\ell} + \gamma^K V^\pi(s_{t+K}) \right) \,\middle|\, s_t = s \right] \\ &\quad = \gamma^K \mathbb{E}_\pi[V_{\text{old}}(s_{t+K}) - V^\pi(s_{t+K}) \mid s_t = s]. \end{aligned}$$

Using $e(x) = V_{\text{old}}(x) - V^{\pi}(x)$ gives

$$\mathbb{E}[Y_t^{(K)} \mid s] - V^{\pi}(s) = \gamma^K \mathbb{E}[e(s_{t+K}) \mid s]. \tag{5.4}$$

The left-hand side is the *conditional mean bias of the target*, not the error of an individual sampled return and not the final fitted critic error. For $K = 1$, it reduces to $\mathbb{E}[r_t + \gamma V_{\text{old}}(s_{t+1}) \mid s] - V^{\pi}(s) = \gamma \mathbb{E}[e(s_{t+1}) \mid s]$. There is no $-e(s)$ term here because this is a return target; a TD residual would additionally subtract $V_{\text{old}}(s_t)$.

Additional observed rewards postpone the point at which the target substitutes a prediction for future experience. If $|e(s)| \le \varepsilon$ everywhere, the bound follows explicitly from

$$\left|\mathbb{E}[Y_t^{(K)} \mid s] - V^{\pi}(s)\right| = \gamma^K \left|\mathbb{E}[e(s_{t+K}) \mid s]\right| \le \gamma^K \mathbb{E}[|e(s_{t+K})| \mid s] \le \gamma^K \varepsilon.$$

5.3 Finite GAE and its critic regression target

Define the TD residual and GAE fragment as

$$\delta_t = r_t + \gamma V_{\text{old}}(s_{t+1}) - V_{\text{old}}(s_t), \qquad \hat{A}_t^{(K)} = \sum_{\ell=0}^{K-1} q^{\ell} \delta_{t+\ell}, \quad q = \gamma\lambda. \tag{5.5}$$

Every fragment here ends with the appropriate next-state bootstrap. The PPO-style critic target is $Y_t^{\text{GAE}} = V_{\text{old}}(s_t) + \hat{A}_t^{(K)}$. Collecting the coefficients of each value prediction gives

$$\begin{aligned} Y_t^{\text{GAE}} = & \sum_{\ell=0}^{K-1} q^{\ell} r_{t+\ell} + \gamma(1-\lambda) \sum_{u=1}^{K-1} q^{u-1} V_{\text{old}}(s_{t+u}) \\ & + \underbrace{\gamma q^{K-1}}_{\beta_K} V_{\text{old}}(s_{t+K}). \end{aligned} \tag{5.6}$$

Equivalently, it is a mixture of bootstrapped n-step returns:

$$Y_t^{\text{GAE}} = (1-\lambda) \sum_{n=1}^{K-1} \lambda^{n-1} Y_t^{(n)} + \lambda^{K-1} Y_t^{(K)}. \tag{5.7}$$

The mixture weights sum to one. Subtracting the true-value version and taking expectations yields the exact target bias

$$\begin{aligned} b_K(s) &= \mathbb{E}[Y_t^{\text{GAE}} \mid s] - V^{\pi}(s) \\ &= \gamma(1-\lambda) \sum_{u=1}^{K-1} q^{u-1} \mathbb{E}[e(s_{t+u}) \mid s] + \gamma q^{K-1} \mathbb{E}[e(s_{t+K}) \mid s]. \end{aligned} \tag{5.8}$$

This result conditions on state and averages the current action under π. For advantage error conditional on (s, a), the corresponding expression also contains $-e(s)$, with future expectations conditional on (s, a). The fixed-policy analysis follows the GAE construction in [1]; it does not pool targets generated by changing policies as if they shared one true value.

6 Quantifying the effect of unequal support

6.1 Endpoint-error sensitivity

Hold the rewards and intermediate predictions fixed and perturb the endpoint prediction by ε. Equation (5.6) gives

$$\boxed{\Delta Y_t = \beta_K \varepsilon, \qquad \beta_K = \gamma q^{K-1}.} \tag{6.1}$$

For the samples at episode position t, the average sensitivity is

$$\bar{\beta}_t = \frac{1}{m}\sum_{j=1}^{m} \gamma q^{K_{j,t}-1}. \tag{6.2}$$

If each fragment endpoint has the same additive prediction perturbation, while its intermediate values remain fixed, the mean target shift is $\bar{\beta}_t\varepsilon$. This is a controlled sensitivity experiment, not the assumption that all states of a single shared critic have this perturbation.

For a complete phase period, Equations (3.5)–(3.6) also give an exact expression for this coefficient:

$$\boxed{\bar{\beta}_t = \frac{\gamma}{L}\left[q^{c_t-1}\frac{1-q^{dh_t}}{1-q^d} + (L-h_t)q^{R_t-1}\right], \quad 0 < q < 1.} \tag{6.3}$$

For incomplete periods, use the period sum and the actual chronological remainder just as in Equation (3.2).

Table 3: Endpoint sensitivity with $\gamma = 0.995$, $\lambda = 0.95$, $M = 500$, $N = 256$, $m = 64$. A perturbation of 10 is in the chosen value units.

t	$\bar{K}_t$	$\bar{\beta}_t$	Mean target shift for $\varepsilon = 10$
1	130.0000	0.065112	0.65112
400	80.6875	0.078992	0.78992
450	46.1250	0.112681	1.12681
499	2.0000	0.940524	9.40524
500	1.0000	0.995000	9.95000

The final position transmits almost the full endpoint prediction error into its regression target. Its endpoint sensitivity is approximately 15.3 times that of the first position, averaged over the phase period. This comparison isolates the sampling structure; Equation (5.8) remains the expression for total bias when intermediate prediction errors are also present.

6.2 Why the support distribution, rather than its mean alone, matters

Define the empirical support-length distribution

$$p_t(k) = \frac{1}{m}\sum_{j=1}^{m} \mathbf{1}\{K_{j,t} = k\}. \tag{6.4}$$

Then $C_t/m = \mathbb{E}_{p_t}[K]$, whereas $\bar{\beta}_t = \gamma\mathbb{E}_{p_t}[q^{K-1}]$. For two length sets $(1, 5)$ and $(3, 3)$, both have cumulative support 6, but

$$\frac{\gamma}{2}(1+q^4) - \gamma q^2 = \frac{\gamma}{2}(1-q^2)^2 > 0. \tag{6.5}$$

Consequently, replacing every K with $\bar{K}_t$ does not give the correct average sensitivity. Indeed, convexity gives

$$\bar{\beta}_t \geq \gamma q^{\bar{K}_t - 1}. \tag{6.6}$$

Short fragments can dominate endpoint sensitivity even when average reward support is large. In particular, the first position's average support 130 does not mean it behaves like a single 130-step estimator.

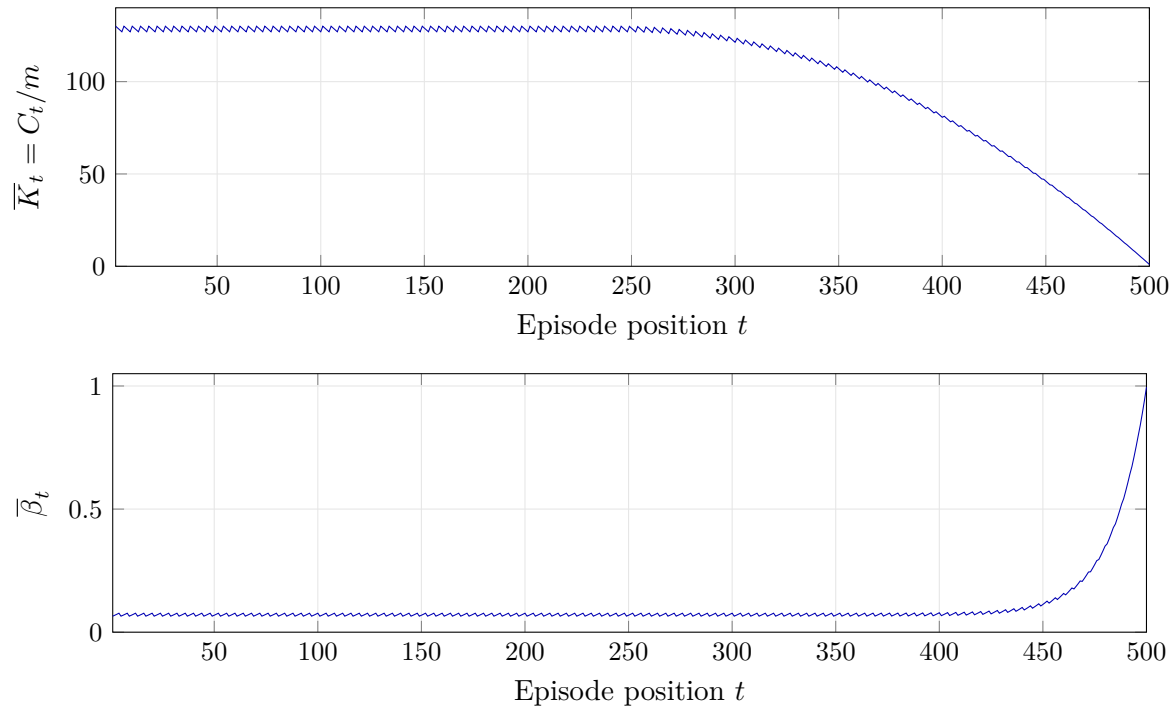


Figure 2: Average observed reward support (top) and average endpoint-error sensitivity (bottom) over 64 episodes. Small oscillations reflect the four residue classes of the rollout phase. Endpoint sensitivity rises sharply near the episode boundary. The bottom curve is a coefficient, not measured critic error.

6.3 A uniform bound on all critic errors

If $|e(s)| \le \varepsilon$, Equation (5.8) implies

$$|b_K(s)| \le \alpha_K \varepsilon, \tag{6.7}$$

where

$$\begin{aligned} \alpha_K &= \gamma(1-\lambda)\frac{1-q^{K-1}}{1-q} + \gamma q^{K-1} \\ &= \frac{\gamma(1-\lambda)}{1-q} + \frac{\gamma\lambda(1-\gamma)}{1-q} q^{K-1}. \end{aligned} \tag{6.8}$$

For the example parameters, $\alpha_1 = 0.995$ and $\alpha_\infty \simeq 0.90868$. Increasing K reduces this uniform bound, but it does not eliminate intermediate-value dependence when $\lambda < 1$. If $e(s) = c$ is constant across all visited states, the target bias is exactly $\alpha_K c$; if $e = 0$, the target bias is zero for every support length. At $\lambda = 1$, $\alpha_K = \gamma^K$; at $\lambda = 0$, $\alpha_K = \gamma$ for every K. These cases distinguish endpoint-specific sensitivity from the total influence of a critic whose errors extend across the entire trajectory.

6.4 From target error to the critic update

For one sample under a squared loss,

$$\ell_t = \tfrac{1}{2}(V_\phi(s_t) - Y_t)^2, \qquad \frac{\partial \ell_t}{\partial V_\phi(s_t)} = V_\phi(s_t) - Y_t. \tag{6.9}$$

Holding the current network fixed, a target perturbation produces

$$\Delta \nabla_\phi \ell_t = -\Delta Y_t \, \nabla_\phi V_\phi(s_t). \tag{6.10}$$

A persistently shifted target repeatedly pushes the fitted value in that direction. Shared network parameters also couple updates at different states. This equation describes the local squared-loss update; value clipping or other optimization modifications can alter the realized gradient.

7 Sample count, coverage, and the remaining error

7.1 More episodes versus longer reward support

For m independent, identically distributed targets at the same state, the sample mean estimator satisfies

$$\mathbb{E}[(\hat{V}(s) - V^{\pi}(s))^2] = b_K(s)^2 + \frac{\mathrm{Var}(Y^{(K)} \mid s)}{m}. \tag{7.1}$$

For independent but nonidentically distributed targets, the corresponding terms are the squared mean bias and $m^{-2}\sum_j \mathrm{Var}(Y_j \mid s)$; correlated targets also add covariance terms. Increasing m reduces sampling noise under the stated independence conditions. It does not automatically remove a systematic target bias. Increasing K changes the target itself, including both bias and variance. For the final position, increasing m changes $C_M = m$ but leaves $\bar{K}_M = 1$ and $\bar{\beta}_M = \gamma$ unchanged.

A useful supplementary statistic is the sum of reward coefficients:

$$W_t = \sum_{j=1}^{m} \frac{1 - q^{K_{j,t}}}{1 - q}, \qquad 0 < q < 1. \tag{7.2}$$

W_t measures total coefficient weight, not an effective independent sample size. Neither C_t nor W_t alone specifies the learned critic's mean squared error.

7.2 Why extending episodes can improve endpoint values

Consider a deterministic chain $s_1 \to \cdots \to s_M \to s_{M+1}$, where training always resets at s_{M+1}. Correct bootstrap retains $Y_M = r_M + \gamma V_{\mathrm{old}}(s_{M+1})$, but there is no observed outgoing transition from s_{M+1} in this dataset. If all observed Bellman equations are satisfied and the boundary value has error c, subtracting the true Bellman equations gives

$$e(s_t) = \gamma e(s_{t+1}) = \boxed{\gamma^{M-t+1} c}. \tag{7.3}$$

Thus small Bellman residuals on the observed chain do not determine the unknown boundary value. Extending episodes makes previously unobserved outgoing transitions available and moves the unconstrained boundary farther away. States already covered elsewhere can share these constraints through learning; what matters is state-transition coverage, not the timestep label by itself.

A rollout boundary is different: the next rollout normally resumes the same episode, so the outgoing data are eventually collected. It limits the current GAE fragment without necessarily creating a persistent coverage gap. An episode reset removes the continuation unless other trajectories visit the relevant states.

8 Implications for the controlled experiment

The direct, quantifiable effect is:

> Late-episode targets have less observed reward support and greater sensitivity to their final value prediction. Correct bootstrap supplies that prediction but does not supply the missing realized rewards. Longer episodes can both extend some fragments and train the predictions on newly observed continuation states.

The following measurements separate these mechanisms:

1. Record $K_{j,t}$, C_t, and $p_t(k)$ by episode position, together with $\bar{\beta}_t$. These are sampling-geometry quantities, not critic-error estimates.

2. Freeze a policy, critic, and normalization statistics; use common long trajectories to compare targets constructed with different rollout/reset cuts. Estimate reference values from independent continuations where feasible. This separates estimator differences from policy changes during training.

3. Measure target error and critic error by position, as well as current, intermediate, and endpoint prediction errors. A larger $\bar{\beta}_t$ matters most where endpoint errors are substantial or systematically signed.

4. Compare training runs with the same physical timestep, reference prefix, reward definition, total interaction budget, and correct timeout bootstrap. Longer episodes change the state distribution: the first H steps occupy a fraction H/M of samples when $H \leq M$, which should be reported.

At fixed rollout length, increasing M does not remove rollout boundaries. For example, in the first episode with $N = 256$, position 500 has support 1 when $M = 500$, but support 13 when $M \geq 512$, because the next rollout boundary is 512. Further extension can improve learned continuation values even when it does not increase that particular sample's immediate support.

For E synchronized environments with identical episode lengths and phases, all environments have the same $K_{j,t}$. Pooling m episodes *per environment* multiplies C_t by E but does not change $\bar{K}_t$ or $\bar{\beta}_t$. Variable episode lengths and early failures require the actual cumulative start index in Equation (2.3); the fixed-length periodic formulas no longer apply.

This document assumes the timeout semantics described in [2]: artificial episode boundaries retain final-state bootstrap, whereas genuine terminal states have zero continuation value. It makes no claim that a particular backend currently enforces these semantics; they are the stipulated conditions of this analysis. All numerical tables use $M = 500$ as a mathematical example, not as a measurement of the actual simulator horizon.

A Reproducing the counts and sensitivity coefficients

The following Python fragment uses only the standard library. It verifies the full-period closed forms against enumeration and prints the values used above.

```
from math import gcd
M, N, m = 500, 256, 64
gamma, lam = 0.995, 0.95
q = gamma * lam
d, L = gcd(M, N), N // gcd(M, N)
for t in range(1, M + 1):
    R = M - t + 1
    K = [min(R, N - ((j*M + t - 1) % N))
         for j in range(m)]
    a = (t - 1) % d
    c = d - a
    h = min(L, max(0, (R-c)//d + 1))
    S = h*c + d*h*(h-1)//2 + (L-h)*R
    beta = gamma/L * (
```

```
        q**(c-1)*(1-q**(d*h))/(1-q**d)
        + (L-h)*q**(R-1))
    assert sum(K) == S  # here m == L
    measured = sum(gamma*q**(k-1) for k in K)/m
    assert abs(measured-beta) < 1e-12
    if t in [1, 100, 256, 257, 400, 450, 499, 500]:
        print(t, sum(K), sum(K)/m, measured)
```

References

[1] J. Schulman, P. Moritz, S. Levine, M. Jordan, and P. Abbeel. *High-Dimensional Continuous Control Using Generalized Advantage Estimation*. ICLR, 2016. arXiv:1506.02438.

[2] F. Pardo, A. Tavakoli, V. Levdik, and P. Kormushev. *Time Limits in Reinforcement Learning*. ICML, PMLR 80:4045–4054, 2018. Paper and proceedings.